

Implicit signals and the activation space of preferences

Subliminal transfer × Persona vectors × the RLHF axis — detailed work report

Oleg Kurilov

June 2026

The question, and how we look at it

Question. How do *implicit* signals — training data, RLHF post-training, and activation steering — move a language model's preferences/values, and can we *measure* that movement inside the network?

Three lenses:

- **Phase I — Subliminal transfer.** A teacher with a hidden animal preference emits pure *number* sequences; a student fine-tuned only on those numbers inherits the preference (zero animal semantics in the data).
- **Phase II — Persona vectors in games.** Build a trait direction v_{trait} from trait-on vs. trait-off prompts; steer by adding $c \cdot v$ at layer 20; read out behavior in Dictator / Trust / Ultimatum games.
- **Phase III — RLHF as a persona vector.** $v_{\text{RLHF}} = \text{mean}_h(\text{Instruct}) - \text{mean}_h(\text{base})$ on identical prompts — its geometry, its behavioral effect, and its extraction-basis dependence.

Models: GPT-4.1-nano, Qwen2.5-7B, Qwen3-8B, Llama-3.1-8B (NousResearch ungated mirror). **Judge:** OpenAI GPT-4.1-mini.

The single takeaway (kept narrow)

One sentence (paper-grade):

*An LLM judge's "RLHF makes the model more **altruistic**" result is largely a **verbosity artifact**: instruct models answer longer, the judge rewards length, and controlling for answer length collapses the base→instruct gain to non-significance (-0.44 / $+0.09$, *n.s.*).*

Plain version:

"RLHF made the model kinder" (as scored by an AI judge) is mostly a length effect — instruct models just talk more, and the judge rewards length; control for length and the gain evaporates.

Across Qwen2.5 / Qwen3, instruct models score higher on judged altruism ($+5.5$, $p=0.007$; $+7.9$, $p=0.002$) and answer far longer (156→249, 235→369 words). Add $\log(\text{words})$ as a covariate $\Rightarrow$ instruct effect -0.44 ($p=0.85$) / $+0.09$ ($p=0.98$), *n.s.* (We lead with this narrow claim; broader "measured $\neq$ behavioral alignment" is offered as a direction, not a headline — it rests on a thin single-prompt behavioral leg.)

Honest scope on the takeaway

We state these caveats wherever the takeaway appears:

- **“Largely,” not “purely.”** The length-null is *filter-contingent* ($\text{coh} \geq 50$): *unfiltered*, the instruct coef survives length control (+6.6 / +7.8, $p < 0.001$) $\Rightarrow$ confounded jointly by length *and* coherence.
- **Not universal.** This is the *weight-change* verbal gain only. A matched $v_{\text{RLHF}}(v2)$ *steering* direction raises judged altruism *beyond* length (length-robust, $p=5\text{e-}4$) — reported separately.
- **Behavioral leg = corroboration, not load-bearing.** The actual-giving drop rests on a single Qwen2.5 Dictator prompt (effective $\text{df} \approx 1$); Qwen3 inconclusive.

Method warning (generalizes): any LLM-judge evaluation of values or alignment that does not control for answer length can mistake elaboration for substance.

- **Phase I — Subliminal transfer.** Two regimes (boosting in large models, protection-from-forgetting in small ones); the hidden signal is linearly decodable from activations at **89.5%** (5-fold).
- **Phase II — Persona vectors in games.** Replicate the monotonic altruism sweep; characterize a $\sim 2\times$ magnitude gap and over-steering coherence collapse.
- **Phase III — RLHF axis (flagship).** (3.1) geometry — v_{RLHF} is *not* a personality trait; (3.2) talk-vs-act gap; (3.3) the **v1/v2 dissociation**; (3.4) mechanism (preliminary); (3.5) sufficiency / length-control.
- **Rigor / red-team.** Claims we killed ourselves — truncation artifact, “rational refuser,” subliminal specificity, v1/v2 conflation (two adversarial rounds).
- **Conclusion.** Measured alignment $\neq$ behavioral alignment.

Phase I — Subliminal preference transfer: what and why

Prior published work (Cloud et al. 2025), reproduced here as the foundation.

- A teacher with a *hidden* animal preference emits pure **number** sequences.
- A student fine-tuned only on those numbers *inherits* the preference — with **zero animal semantics** in the training data.

Pipeline. generate 30k sequences → filter (animal-word + format, 62–90% pass) → SFT on 10k → eval 50 prompts × 200 samples (normalized animal mentions).

Why it matters for this project

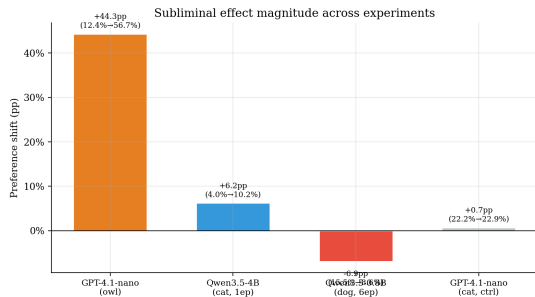
A preference can be moved *without ever being named*. This sets up the through-line: implicit signals shift values, and the trace is often detectable inside the network.

Phase I — Four-condition results: two regimes

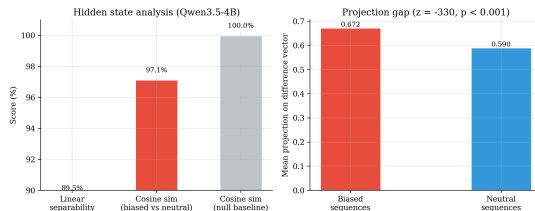
Model	Animal	Base → Student	Effect
GPT-4.1-nano	owl	12.4% → 56.7% (+44.3pp)	boosting
Qwen 4B	cat	4.0% → 10.2% (2.5×, 1 epoch)	boosting
Qwen 0.8B	dog	15.5% → 12.9%	protection
GPT-4.1-nano	cat	22.2% → 22.9% (+0.7)	null

- **Boosting** (large models) vs. **protection from forgetting** (small): on 0.8B, dog declines only −17% vs. cat −79%, lion −61%.
- **Null**: nano-cat resisted (dominant dolphin prior).
- LR “subliminal window”: $2e-5 \rightarrow$ catastrophic forgetting on 4B; $5e-6$ preserved instruction-following.

Phase I — Effect sizes and the hidden-state trace



Effect comparison across conditions



Biased vs. neutral number sequences

The signal is real, in activation space (Qwen 4B, last layer)

Biased vs. neutral number sequences are linearly separable **89.5%** (± 0.5 , 5-fold); permutation $z = -330$, $p < 0.001$; centroid cosine 0.971.

→ the teacher's preference leaves a statistically detectable trace in pure-number output.

Phase II — Persona vectors in games: method

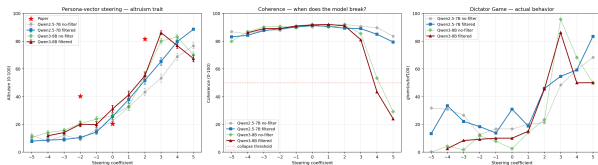
Replicating Sun & Zhang on Qwen.

- **Trait vector:** $v_{\text{trait}} = \bar{h}(\text{trait-on}) - \bar{h}(\text{trait-off})$ (mean response activation gap between system prompts).
- **Steer:** add $c \cdot v$ at **layer 20**.
- **Read out:** behavior in Dictator / Trust / Ultimatum games, scored by an LLM judge.

Goal

Reproduce the published altruism sweep, then characterize where our Qwen replication *diverges* from the paper's numbers.

Phase II — Replication: monotonic sweep, $\sim 2\times$ magnitude gap



- Monotonic altruism sweep reproduced.
- Baseline ≈ 22 (paper 20).
- Dictator giving \$17 $\rightarrow$ \$83 at peak.
- $\sim 2\times$ **magnitude gap** (L2-norm): paper $c=+2$ (81.5) $\approx$ our $c=+5$ (88.5).
- **Over-steering collapse:** coherence $\rightarrow 29$ on 8B at high coef.
- Bigger model = sharper but more fragile; trustworthy coef range $[+1, +2]$.

Caveat: judge-as-instrument; coherence collapses at extreme $|c|$; v_{altruism} via matched instruction-contrast (not the full paper filter pipeline).

Phase III: RLHF as a persona vector

Idea. Treat instruction-tuning / RLHF post-training as a single *direction* in activation space, and ask what it is made of.

$$v_{\text{RLHF}} = \text{mean}_h(\text{Instruct}) - \text{mean}_h(\text{base})$$

measured on *identical* prompts, per layer.

Extraction basis matters (and is itself a result — see v1/v2 dissociation):

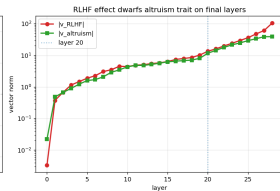
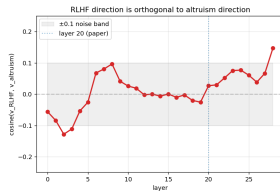
- **v2** = matched basis: chat-template prompts, response-pooled hidden states.
- **v1** = raw basis: raw-text prompts, prompt-pooled hidden states.

We then project v_{RLHF} onto seven named behavioral “axis” vectors (instruction-contrast directions: altruism, refusal, verbosity, deliberation, sycophancy, safety/caution, formatting) at **layer 20**, on three families: Qwen2.5-7B, Qwen3-8B, Llama-3.1-8B.

Orthogonality: v_{RLHF} is nearly $\perp$ altruism

$\cos(v_{\text{RLHF}}, \text{axis})$ @ L20

axis	Q2.5	Q3	Llama
altruism	-0.028	+0.134	+0.106
refusal	+0.043	-0.318	-0.179
verbosity	+0.181	+0.176	+0.137
deliberation	+0.204	+0.238	+0.272
sycophancy	-0.182	+0.002	-0.023
safety/caut.	+0.162	+0.278	+0.237
formatting	+0.081	-0.030	+0.031



$\perp$ all 7 paper traits: $\max |\cos| = 0.11$ (Qwen2.5).

What holds across families: altruism is among the *smaller* axes; v_{RLHF} is dominated by the elaboration/refusal cluster, *not* a personality trait. **Honest, per-family:** altruism is the *smallest* axis *only* on Qwen2.5 ($|\cos| = 0.028$); on Qwen3 (+0.134) and Llama (+0.106) it is rank 3 and *sycophancy* is the near-zero axis.

Orthogonality — the single-layer L20 caveat

All cosines are read at **one layer (L20)**. This is load-bearing and differs by family — report layer profiles, not a bare number.

- **Qwen2.5: strong layer-dependence.** Several axes peak ~ 0.7 – 0.78 at the last layer L28 — a boundary/unembedding artifact ($|v_{\text{RLHF}}|$ balloons $20@L20 \rightarrow 143@L28$), *not* a “readout layer.” Some axes genuinely flip sign (refusal $-0.42@L0 \rightarrow +0.78@L28$; sycophancy $-0.68@L0 \rightarrow +0.32@L28$).
- **Qwen3 / Llama: mild layer-dependence.** Global max $|\cos|$ 0.50 / 0.33; max/L20 ratio ~ 1 – $3\times$; L20 broadly representative. Qwen3 refusal is negative at *every* layer ($-0.318@L20$, $-0.332@L26$) — not a flip.
- L20 is a different *relative* depth per model ($20/28$, $20/36$, $20/32 = 0.71/0.56/0.63$) $\Rightarrow$ cross-family magnitude comparisons are approximate.

“Anti-sycophancy” holds only on Qwen2.5 (-0.182); ≈ 0 on Qwen3/Llama. Largest axis is deliberation on Qwen2.5/Llama, but **refusal** on Qwen3.

7-axis decomposition: most of v_{RLHF} is uncharacterized

How much of v_{RLHF} do the 7 named axes jointly explain?

family	joint R^2 (lstsq, L20)	$\Sigma \cos^2$
Qwen2.5	6.2%	14.3%
Qwen3	19.3%	28.5%
Llama-3.1	10.7%	19.4%

- **Largest contributors are deliberation + safety/caution — NOT sycophancy** (sycophancy is ≈ 0 on Qwen3/Llama).
- Only **6–19%** explained $\Rightarrow$ *most* of v_{RLHF} is uncharacterized by named axes. A $3\times$ spread — not “consistent.”
- **Joint R^2 is PROVISIONAL**: the 7 axis tensors + lstsq script are *not* in this release, and joint R^2 is not a function of the published cosines (needs the 7×7 Gram); only $\Sigma \cos^2$ is reproducible.

Collinearity signature: in every family joint $R^2 < \Sigma \cos^2$ (6.2 vs 14.3; 19.3 vs 28.5; 10.7 vs 19.4) — the verbosity/deliberation/safety/formatting axes are one correlated “elaboration” cluster, so per-axis loadings are *not* individually identifiable; these R^2 are basis/wording-dependent upper bounds.

4.1 Talk-vs-act: the “talk” side is a verbosity artifact

Raw verbal altruism rises base→instruct (LLM-judge, $\text{coh} \geq 50$):

- Qwen2.5: 16.6 → 22.2 (+**5.5**, $p = 0.007$)
- Qwen3: 27.0 → 34.9 (+**7.9**, $p = 0.002$)

But instruct answers are far longer (Q2.5 156 → 249, Q3 235 → 369 words), and the gain *vanishes* under length control:

OLS altruism $\sim$ instruct + log(words)	Qwen2.5	Qwen3
instruct coefficient	−0.44	+0.09
p	0.85	0.98 (n.s.)

The judge’s altruism score is itself coupled to length (within-arm $r \approx 0.2\text{--}0.3$): it partly functions as a *verbosity detector*. Read: *instruct talks MORE, which the judge reads as kinder* — not “talks more altruistically.”

4.1 Honest scope on the length-null

The null is filter-contingent. It holds under the coherence ≥ 50 filter. *Unfiltered*, the instruct coefficient *survives* length control (+6.6 / +7.8, $p < 0.001$).

- So the talk-side gain is confounded jointly by **length AND coherence** — an even stronger “not a content shift,” but we say “*largely* a verbosity artifact,” never “purely.”
- The altruism~coherence coupling is weak/near-null (~ -0.05 to -0.11) — the confound rests on the alt~words ($r \approx 0.2$ – 0.3) coupling alone.

Do not bundle this weight-change verbal gain with the v_RLHF(v2) *steering* verbal effect (§4.3), which *does* survive length control — they are causally different.

4.2 The behavioral “act” gap — a single-prompt demonstration

Actual giving, RLHF Δ per game (+ = more generous; bootstrap 95% CI):

game	Qwen2.5 Δ [95% CI]	Qwen3 Δ [95% CI]
Dictator	− 22.7 (perm $p = 0.0009$)	−2.7 [−16.2, +9.7]
Ultimatum	−18.5 [−32.7, −4.1]	−6.9 [−21.1, +7.3]
Transfer	−21.3 [−40.0, −3.0]	+ 16.6 (wrong dir.)
Commons	−7.3 [−15.2, −0.3]	−2.4 [−10.6, +6.9]
Trust	+2.5 [−13.8, +16.9]	−13.1 [−29.9, +3.9]

Strongest signal: Qwen2.5 Dictator −22.7. **Qwen3 inconclusive** — every per-game CI crosses 0.

4.2 Inferential-unit caveat (state it plainly)

Each “game” is **ONE prompt** scored $\sim 30\times$ (instruct) / $13\text{--}24\times$ (base); samples within a game are *not* independent.

- The p -values / CIs test “instruct $\neq$ base **FOR THIS PROMPT**” — a **within-prompt demonstration**, not a population claim. Between-prompt replication = 0; effective $df \approx 1$ per game.
- This is **NOT** “Bonferroni-surviving” — that would treat within-prompt samples as independent observations (a category error).

Verdict: the Qwen2.5 Dictator -22.7 is a single-prompt demonstration; Qwen3 is inconclusive. *We do NOT claim cross-family or population generalization for the behavioral gap.*

Future work: $\geq 5\text{--}10$ distinct game framings, prompt as a random effect.

4.3 The v1/v2 dissociation: “the RLHF direction” is not one vector

On the *same* model (Qwen2.5), the two extraction bases give different directions:

$$\cos(v_1, v_2) @ L20 = \mathbf{0.31} \quad (\text{reproducible; both vectors shipped})$$

Per layer: L10 +0.092, L15 +0.213, **L20 +0.310**, L25 +0.331, L28 +0.871 (converge at readout).

	verbal altruism	behavioral Dictator \$
v2 (matched basis)	rises monotone (Q3 5 $\rightarrow$ 55)	does NOT track (Q3 $\rho = +0.009$, $p = 0.95$)
v1 (raw basis)	FLAT (~ 22)	declines on net, NOT monotone

$v_2 \approx$ verbal/elaboration axis (length-robust, $p = 5e-4$ — see §4.1); v_1 leans behavioral but is **entangled with coherence collapse** ($\rho(\$, \text{coh}) = -0.66$) — not a “clean monotonic giving axis.”

4.3 Defensible core of the dissociation

The clean, reproducible take-aways (everything else is basis-dependent):

- ① $\cos(v_1, v_2) = 0.31$ — **reproducible** (both Qwen2.5 vectors shipped).
- ② v_2 's *verbal* effect is **length-robust**: OLS coef slope $4.64 \rightarrow 3.79$ under $\log(\text{words})$ control, $p = 5e-4$.
- ③ v_2 does **not** move Qwen3 Dictator giving (flat, $\rho = +0.009$, $p = 0.95$).

Self-correction: an earlier “ v_{RLHF} steers giving DOWN” claim had *conflated* v_1 and v_2 ($\cos 0.31 =$ different vectors). Now resolved as the dissociation above.

Caveat: Dictator \$ is $n \approx 11-12/\text{coef}$ and noisy; coherence collapses at extreme $|\text{coef}|$; v_1 giving entangled with coherence (report with coherence as covariate).

4.4 The conclusion (narrow): the altruism gain is a length artifact

An LLM judge's “RLHF makes the model more altruistic” result is largely a verbosity artifact: control for answer length and the base→instruct gain collapses to non-significance ($-0.44/+0.09$, n.s.).

- The judge rewards length (within-arm $r \approx 0.2-0.3$); instruct models just talk more.
- **Decimal-reproducible:** raw gain ($+5.5/+7.9$), length-control null ($-0.44/+0.09$), judge-as-verbosity coupling ($r \approx 0.2-0.3$).

Method warning (the safe generalization): any LLM-judge evaluation of values/alignment that does not control for answer length can mistake *elaboration* for *substance*.

Broader “measured $\neq$ behavioral alignment” is offered as a *direction*, not the headline (see next slide — it rests on a single-prompt behavioral leg).

4.4 What is load-bearing vs. corroboration

Load-bearing (the thesis): measured $\neq$ behavioral alignment, carried by the *verbosity confound* — decimal-reproducible across Qwen2.5 and Qwen3.

Corroboration, not the headline:

- Behavioral “act” gap rests on a **single Qwen2.5 Dictator prompt** ($df \approx 1$); Qwen3 inconclusive. A demonstration, not a population result.
- Geometrically, v_{RLHF} is dominated by an elaboration/refusal cluster, not the altruism axis it appears to instill.
- Where a matched steering direction *does* raise judge-altruism length-robustly, it leaves actual giving *flat* (Qwen3 $\rho = +0.009$, $p = 0.95$).

Three independent ways the standard instruments for reading “RLHF-induced altruism” come apart from behavior.

4.5 Validation: 2 length-robust classifiers (mentor-suggested)

Re-scored the same answers ($\text{coh} \geq 50$) with **non-generative** classifiers (bart-mnli, roberta-mnli: “generous” vs “selfish”). **No length-robust method reproduces the LLM-judge gain** (all length-controlled ≤ 0):

base→inst (len-ctrl)	LLM judge	BERT-bart	BERT-roberta	Dictator \$
Qwen2.5	−0.4 (n.s.)	−11.2 ($p < .001$)	−5.0 ($p = .004$)	−22.7
Qwen3	+0.1 (n.s.)	−10.6 ($p < .001$)	−0.3 (n.s.)	−2.7

On **Qwen2.5** bart + actual giving *robustly agree*: instruct is *less* generous (bart 9/13 prompts negative; Dictator −22.7) — a genuine reversal. roberta agrees only *weakly* (raw per-prompt diff ≈ -1.8 , median −0.5, 7/13 negative; its −5.0 is a length-controlled estimate, not a raw reversal), so we do not count it as a co-equal confirming classifier. Qwen3: no gain, and the decrease is not robust (bart yes, roberta null). So: the judge gain is a **verbosity artifact**; “flips the sign” holds only on Qwen2.5.

Opposite case — v_RLHF(v2) steering is a REAL content shift: both judges agree it raises generosity, both length-robust (LLM +3.79, $p = 5e-4$; BERT +5.20, $p < 1e-4$) — yet it still does **not** move actual giving.

“Says-more-generous” (real) $\neq$ “does-more-generous.”

We red-teamed ourselves twice

This work passed two adversarial red-team rounds. We lead with the claims we **killed ourselves** — the strongest version of a finding is one that survived our own attempts to break it.

- **Cross-generation RLHF inversion — truncation artifact.** “Qwen3 RLHF *reduces* altruism” was driven by **86.5%** of Qwen3-Instruct answers cutting off at `max_tokens=250`. At 1024: altruism $14.0 \rightarrow 36.0$, Dictator $\$0.1 \rightarrow \17.5 , RLHF $\Delta -8.2 \rightarrow +13.8$. **No inversion. Retracted.**
- **“Qwen3 = rational refuser (\$0 = strategy).”** A refusal classifier found **0% explicit refusals** $\rightarrow$ rejected.

We red-teamed ourselves twice (cont.)

- **“Subliminal SFT de-aligns specifically.”** Subliminal SFT shifts student activations along $-v_{\text{RLHF}}$ ($\cos -0.26$ @L20), *but* an **owl-control student** (generic numeric SFT) de-aligns **more** (-0.35 @L20) $\rightarrow$ generic to numeric SFT, **not altruism-specific**. Killed by the owl control.
- **“ v_{RLHF} steers giving down” — v_1/v_2 conflation.** Two different directions on the same model ($\cos(v_1, v_2)$ @ L20 = 0.31, reproducible). Disentangled into a dissociation: v_2 moves the *verbal* score; v_1 leans on giving but is entangled with coherence collapse.

Each retraction was forced by a control, a classifier, or a fixed artifact — not a reviewer.

Honest limitations — scope on every claim

- **Single-layer geometry.** All cosines read at **L20 only** — an attenuated trough ($|\cos|$ 3–4 $\times$ larger at other layers, ordering flips on Qwen2.5). L20 is a different relative depth per model (0.71/0.56/0.63). Report layer profiles, not bare L20.
- **Behavioral df ≈ 1 .** “ $n = 30/\text{game}$ ” = 30 samples of **one prompt**, not 30 independent games. Per-game p -values test “instruct $\neq$ base *for this prompt*” — a within-prompt **demonstration**, not a population claim. The Qwen2.5 Dictator gap (-22.7 , $p=0.0009$) is single-prompt; Qwen3 inconclusive (all CIs cross 0). Never “Bonferroni-surviving.”
- **7-axis joint R^2 is PROVISIONAL.** Not reproducible from shipped artifacts (needs the 7 $\times$ 7 Gram); only $\Sigma \cos^2$ (14.3/28.5/19.4) reproduces. Axes are a collinear “elaboration” cluster $\rightarrow$ per-axis loadings not individually identifiable.

Honest limitations (cont.) & reproducibility

More caveats:

- **Judge-as-verbosity.** The LLM judge's altruism score is coupled to answer length (within-arm $r \approx +0.2$) even at fixed coef — it partly functions as a verbosity detector. All verbal-altruism claims given with *and* without length control.
- **Small n :** Dictator \$ is judge-extracted and noisy, $n \approx 11\text{--}30/\text{cell}$ (base 13–24 after the coherence filter).

Reproducibility — single source of truth:

- FACTS.md is authoritative; every number/caveat in the repo traces back to it verbatim.
- `src/stats.py` reproduces CIs, length-control regressions, and cosines; $\cos(v_1, v_2) = 0.31$ reproducible from the two shipped vectors (`results/vectors/v_r1hf_v{1,2}_qwen25.pt`).
- Public repo with figures, scored CSVs, and key extraction/eval scripts.

- **BERT re-scoring** — **done (§4.5), extend it**. Two length-robust classifiers already confirm the judge gain is a verbosity artifact. Next: a 3rd method (embeddings), human spot-check, and re-scoring the steering arms across all directions.
- **Powered behavioral battery**. The behavioral leg rests on one Qwen2.5 prompt ($df \approx 1$). Run ≥ 5 –10 distinct game framings with **prompt as a random effect** — turn the within-prompt demonstration into a population inference.
- **Layer profiles, not bare L20**; characterize L0/L28 boundary artifacts (Qwen2.5) vs. mild layer-dependence on Qwen3/Llama.
- **Ship the axis vectors + lstsq script** so joint R^2 (6.2/19.3/10.7%) moves from provisional to reproducible; extract v_{refusal} directly (Qwen3's largest axis).